FOOD REMEDY API

RE024 – Context-Aware Personalised Food Recommendations

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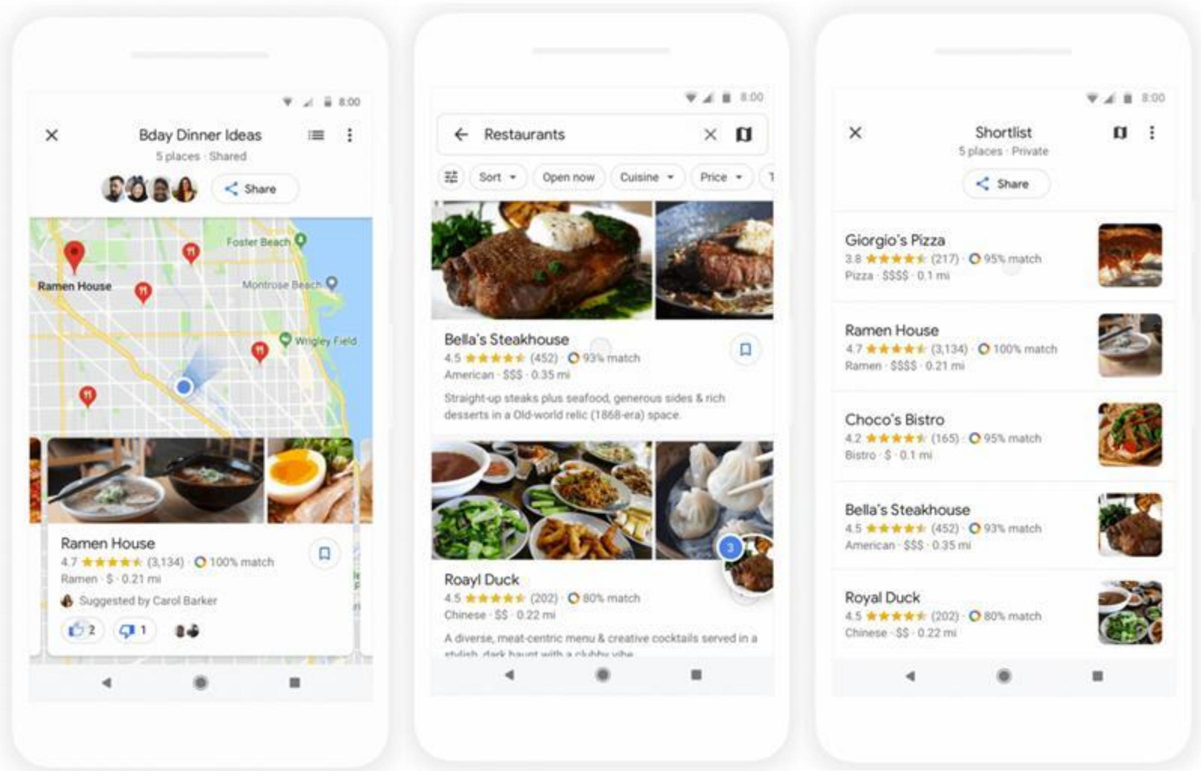
Introduction

Context-aware personalized food recommendations use real-time data to suggest food that fits the user's current situation, not just their saved preferences. In a food-health app, this means that the system can take into account the time of day, where you are, how active you are, your dietary goals, any allergies you have, your mood, the weather, and how many calories you have left before suggesting a meal or product.

How context-aware systems work

Context factor	**How it improves recommendations**
Time of day	Suggests breakfast, lunch, dinner, snacks, or late-night options
Location	Recommends nearby supermarkets, restaurants, or products available locally
User activity	Suggests higher-energy meals after exercise or lighter meals during low activity
Health profile	Filters based on allergies, diet type, goals, and restrictions
Real-time progress	Uses remaining calories, macros, hydration, or logged meals

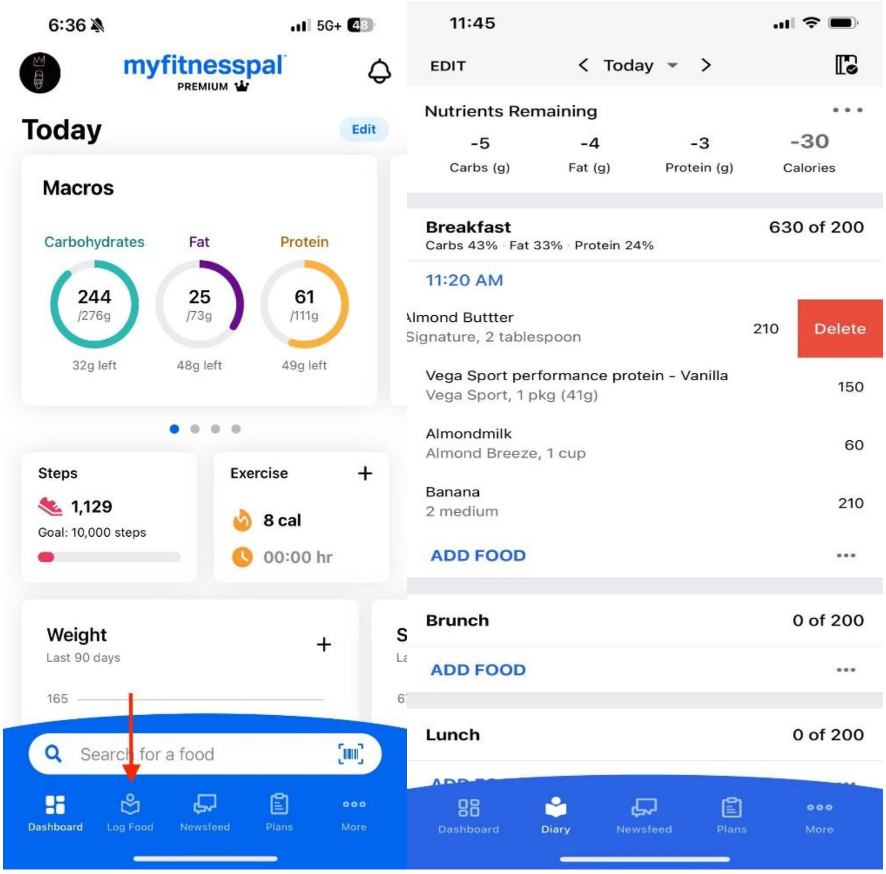
For instance, MyFitnessPal's newer features are all about meal planning, keeping track of meals, macros, calorie goals, and personalized nutrition help based on user data. Google Maps uses your location, the time of day, nearby restaurants, and how busy the area is to give you timely local suggestions.

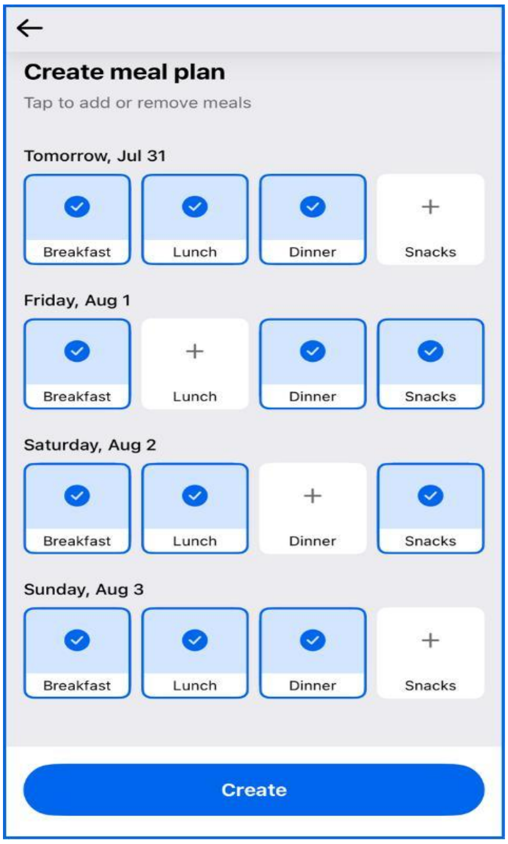


Platform comparison

Platform	**Context used**	**Example use**
Google Maps	Location, opening hours, distance, busyness, user interests	Recommends nearby restaurants that are open now
MyFitnessPal	Meal time, calories, macros, user goals, logged data	Suggests meals based on nutrition targets
Yelp	User preferences, dietary restrictions, reviews, local business data	Personalised restaurant results based on lifestyle and food preferences
DoorDash / delivery apps	Location, time, previous orders, cuisine preference	Suggests nearby meals suitable for lunch/dinner
Food Remedy API	Mood, diet profile, allergens, time, activity, health goals	Could recommend healthier products or meals during shopping

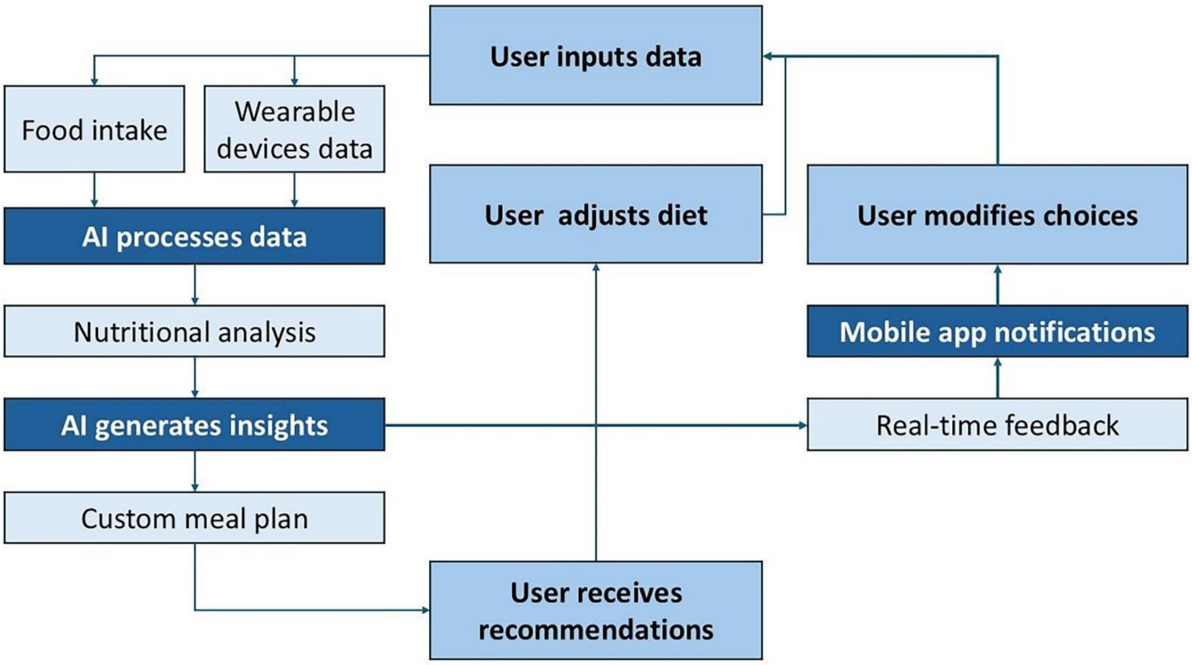
Yelp has also started to give more personalized recommendations based on things like dietary restrictions and lifestyle choices. Recent AI features in local recommendation apps also show that platforms are moving toward real-time and conversational recommendation support.





Real-time inputs and accuracy

Real-time inputs make recommendations more accurate because the system knows what the user needs right then and there. For instance, a person who just worked out and is now at the grocery store may need a high-protein snack, but that same person may need a lighter dinner option at night. Location-aware systems can also get rid of suggestions that aren't useful by only showing options that are nearby or available. Studies on restaurant recommendation systems indicate that location-based filtering enhances relevance by aligning suggestions with geographical limitations.



Key patterns found

The dominant pattern is that modern recommendation systems leverage personal data and the current context. Static preferences such as "vegetarian" or "low sugar" are useful, but real-time context makes the recommendation more pragmatic. The strongest systems are hybrid: user profile + current situation + product/restaurant data + feedback history.

Suggested visual placement

Add Google Maps visual to "Location-driven recommendations" section. Move the MyFitnessPal meal planner visual to the "Time-based and goal-based recommendations" section. Move the meal selection visual near the "Real-time inputs" section to illustrate how the food choices are adjusted based on meal type.

Conclusion

Context-aware personalised food recommendations are important as they make food suggestions more relevant, timely and useful. The system can vary what it suggests to eat each time based on the user's location, the time of day, what they have already eaten, their health goals and their current activity. This approach could also be applied for the Food Remedy API to improve user engagement by providing real-life suggestions during actual shopping or eating situations.